with a chip that differentiates the ultimate product from the other products in the market. Uniqueness is something we look for in any product we buy. Hence it becomes important for all electronic goods to provide something unique, something that is an improvement over the previous model and a characteristic that sets it apart from other products in the market. More often than not, the chip becomes that differentiator. And hence, the importance of the chip designer.

Three paths to choose

If you were to pursue a career in this field you could choose to be a part of three different industries:

- Chip design and development
- Embedded product design industry
- Electronic design automation tools

Currently over 82% per cent of embedded design engineers are employed in embedded software, followed by Chip Design and EDA at 18%.

What are the required skill sets?

Basic requirement is a degree in electronics communication engineering. During interviews, firms look for a good sense of engineering with a sound grounding in mathematics, physics and electronics. Knowledge of different languages is not a must.

A number of courses have come up in design which teach the tools (VHDL, EDA, etc.) and taking them would be a bonus.

Constant improvement of skills and knowledge is a must as the design tools and processes keep evolving.

What is the required mind set?

- **Ability to deal with a level of abstraction:** The science and craft of design is evolving and becoming more complex. Practitioners need to hone their skills over the years.
- **Get a kick out of technology:** Chips are at the root of a lot of product innovations that taking place. Given the amount of effort and time involved in seeing the final product, a person should inherently be in love with technology to keep him going.
- **Patience:** It takes over a year and a half before freshers can start contributing productively to various projects. So don't lose heart early on.

What are the future prospects?

Compared to other engineering fields, for example the mechanical industry, the chip industry is more cyclical in nature. If one is making a career decision, one shouldn't be swayed by the current outlook — the industry has always bounced back from recessionary periods with renewed avenues for growth.

Currently, the industry is looking at alternate energy and MEEMS (Micro Electronics Electromechanical), for example the Nintendo Wii, as the new growth areas.

Sumit Sharma is a 26 year old design engineer, currently working with Samsung India. He has been a part of the embedded design industry for the past four years now. He started his career with Infosys in the product engineering department, and after working for three years changed company to grow vertically. Here's what he had to say about his job.